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CPCS-214 Computer Architecture and Organization

08 - Operands of Computer Hardware

Introduction

- Unlike programs in high-level languages, operands of arithmetic instructions are restricted
- They must be from a limited number of special locations built directly in HW called *registers*

Registers

- Primitives used in HW design that are also visible to programmer when computer is completed
- Size of a register in MIPS architecture is 32 bits

32- bit Register

1100 1000 1111 0011 0101 1010 1110 0111

Data contents

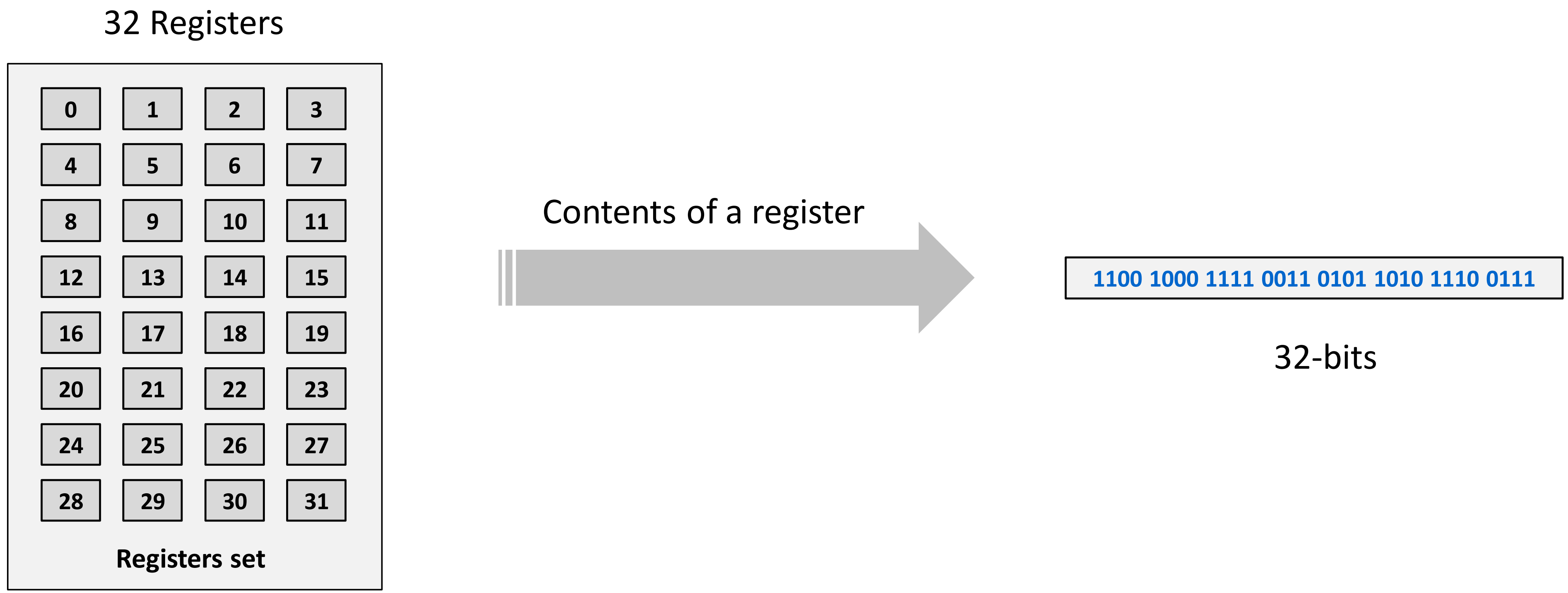
Word

- Groups of 32 bits occur so frequently that they are given the name word in MIPS architecture
  - Natural unit of access in a computer
  - Corresponds to the size of a register in the MIPS architecture

Variables and Registers

- Variables are used in programming language
- One major difference between variables and registers is the limited number of registers
  - Typically 32 on current computers
- In symbolic representation of MIPS language, one restriction is
  - The three operands of MIPS arithmetic instructions must each be chosen from one of 32, 32-bit registers
  - Variable-Register (V-R) mapping (association)

Registers Set (File)



Naming Registers

- We could simply write instructions using numbers for registers, from 0 to 31
- MIPS convention is to use two-character names following a $ to represent a register
- Will use
  - $s0, $s1, . . . for registers that correspond to variables in C and Java
  - $t0, $t1, . . . for temporary registers needed to compile programs

Design Principle 2

- Smaller is faster
  - Very large number of registers may increase clock cycle time because it takes electronic signals longer when they must travel farther
  - Another reason for not using more than 32 is number of bits (5 bits) it would take in the instruction format

Performance

- Registers play central role in HW construction
- Effective use of registers is critical to program performance

Variables-Registers Association

- A variable must be associated to any of the registers
- It is the compiler’s job to associate program variables with registers

Example

- Take the assignment statement:

```
f = (g + h) - (i + j);
```

Variables f, g, h, i, and j are assigned to the registers $s0, $s1, $s2, $s3, and $s4, respectively.

What is the compiled MIPS code?

Example (cont.)

- Compiled program is very similar to prior example, except we replace variables with register names plus two temporary registers, $t0 and $t1:

```
add $t0,$s1,$s2    # register $t0 contains g + h
add $t1,$s3,$s4    # register $t1 contains i + j
sub $s0,$t0,$t1    # f gets $t0 – $t1, which is (g + h) - (i + j)
```